

CURRICULUM VITAE

Part A. PERSONAL INFORMATION

CV date 18/03/2026

First name	Orly Enrique		
Family name	Apolo Apolo		
Gender (*)	Male	Birth date (dd/mm/yyyy)	30/10/1991
Social Security, Passport, ID number	26824167A		
e-mail	enrique.apolo@irta.cat	URL Web:	https://eapolo-lab.github.io/research/
Open Researcher and Contributor ID (ORCID) (*)	0000-0001-8311-0579		

A.1. Current position

Position	Researcher (IRTA Consolida-Research Staff Consolidation Program)		
Initial date	01/10/2025		
Institution	Institut de Recerca i Tecnologia Agroalimentàries (IRTA)		
Department/Center	Postharvest program/Fruitcentre		
Country	Spain	Teleph. number	+34 622-870-120
Key words	agriculture; remote-sensing; artificial intelligence; crops; data fusion; hyperspectral; postharvest; robotics; digitalization		

A.2: Previous professional status (including breaks in research career, according to what is indicated in the call, indicate total months)

Period	Position/Institution/Country/Interruption cause
2024-2025	Research / KU Leuven / Belgium
2023-2023	Postdoctoral researcher / Hasselt University / Belgium
2022-2023	Postdoctoral researcher / Ghent University / Belgium
2018-2022	PIF Fellow / University of Seville / Spain
2018-2018	Research internship contract / University of Seville / Spain

A.3. Education

PhD, Licensed, Graduate	University/Country	Year
Ph.D., Cum Laude, with International Distinction	University of Seville / Spain	2022
M.Sc., in Land Management Instruments and Intervention Techniques	University of Seville / Spain	2017
M.Sc., in Crop Protection	University of Seville / Spain	2017
B.Sc., in Agriculture Engineer	University of Seville / Spain	2016

Part B. CV SUMMARY

My **journey toward scientific independence** began at the University of Seville, where I earned my B.Sc. in Agricultural Engineering (2016) with an **Outstanding Performance Award**. An internship at IFAPA, leading my **first peer-reviewed article**, showed me the transformative potential of digital technologies in agriculture and shaped the direction of my career. In 2017, I completed **two Master's degrees with distinction** and **secured an early-career grant** that allowed me to join a pioneering research group (**AGR-278**) in digital agriculture. These experiences strengthened my passion for developing tools for farmers, especially those related to artificial intelligence (AI). A **PIF Fellowship enabled me to pursue my PhD** where I integrated drones, sensors, and satellite data to create smart tools for agriculture, boosting yield prediction, monitoring crop water stress in real time, and accelerating phenotyping in cereals and maize. This work earned me the **Corteva Chair Award 2024 for the Best Doctoral Thesis** in precision agriculture, chosen from theses defended in Spain and Portugal between 2022-2023. A **research stay at Wageningen University & Research** strengthened my international embeddedness and expanded my expertise in UAV-based modelling in orchard systems and AI. Before defending my thesis, I joined **Ghent University** to work on hyperspectral imaging and AI for disease detection and later continued my postdoctoral training at the **Data Science Institute of Hasselt University and the Flanders AI Academy (VAIA)**, where I strengthened my skills in AI, statistics and methodological innovation, and learned to transfer academic AI solutions to industry. At **KU Leuven**, my research focuses on advancing AI for Earth observation through conditional generative adversarial networks that integrate Sentinel-2 imagery with climate reanalysis

data to model vegetation dynamics. Currently, at IRTA, I am developing AI-driven sensor fusion approaches to improve post-harvest decision-making.

Throughout my career, I have been motivated **by creating tangible benefits for society** through technological innovation. I have contributed to applied projects in robotics, remote sensing, AI and sensor networks, developing practical solutions that support farmers, cooperatives and agri-food companies. My training at VAIA and my role as **Chief AI Consultant for the ag-tech startup Planticus** gave me hands-on experience in technology transfer. This later inspired me to found **Agrocoding**, my own initiative (2023-2024) focused on agricultural digitalization and AI. I have also engaged actively in outreach through podcasts (“**Agricultor Digital**” and “**Science Talk with Naz**”), professional training and dissemination events, helping to bring digital agriculture closer to practitioners and end-users. A central focus of my career has been **my contributions to the training and development of young researchers**. I have supervised 10 MSc theses, 6 BSc theses and 1 PhD dissertation (**Dr. Jaime Rodríguez Vázquez**), and frequently serve as a **jury member for MSc and PhD defences at the Escuela Colombiana de Ingeniería Julio Garavito**. I have delivered more than **260 hours of official Bachelor’s and Master’s teaching and 167 hours of lifelong-learning instruction**, including contributions to the Master of Bioscience Engineering at KU Leuven and the **Máster en Agricultura Digital e Innovación Agroalimentaria at US**, one of the first programs in Southern Europe dedicated to agricultural digitalization. As part of the **REGIONS4FOOD Interreg Europe project**, I trained academic staff at the **Hungarian University of Agricultural and Life Sciences**, helping to build international capacity. I have also worked with the **Universidad de las Américas and the Universidad de Los Lagos (Chile)**, giving invited lectures and supervising undergraduate students. **Other relevant merits** include my role as **reviewer** for Q1 journals in as **Guest Editor of a Special Issue on AI-based yield estimation**. I have published **19 peer-reviewed articles (673 Scopus citations (h-index 9))**, a book, a book chapter, and given **16 international oral presentations**. Despite the constraints that non-permanent researchers face when accessing competitive projects, I have contributed to **11 research projects and 3 industry contracts**. My strong track record in applied and fundamental research, together with proven leadership, mentoring, and dissemination skills.

Part C. RELEVANT MERITS

C.1. Publications

Abbreviations: FWCI: Field-Weighted Citation Impact; CP: Author Position/Total Authors; JQ: JIF Quartile; C: Citations according to Scopus.

- [1] Almoujahed, M. B.; **Apolo-Apolo, O. E.**; Alhussein, M.; Kazlauskas, M.; Kriauciūnienė, Z.; Šarauskis, E.; Mouazen, A. M. **2025**. Deoxynivalenol prediction and spatial mapping in wheat based on online hyperspectral imagery scanning. *Smart Agricultural Technology*, 11, 100947. Doi: 10.1016/j.atech.2025.100947. **FWCI: 2.81; CP: 2/7; JQ: Q1; C: 2.**
- [2] Rodríguez-Vázquez, J.N.; **Apolo-Apolo, O.E.**; Martínez-Moreno, F.; Sánchez-Fernández, L.; Pérez-Ruiz, M. **2025**. Monitoring Leaf Rust and Yellow Rust in Wheat with 3D LiDAR Sensing. *Remote Sensing*, 17, 1005. Doi:10.3390/rs17061005. **FWCI: 1.32; CP: 2/5; JQ: Q1; C: 1.**
- [3] Almoujahed, M. B.; **Apolo-Apolo, O. E.**; Whetton, R. L.; Kazlauskas, M.; Kriauciūnienė, Z.; Šarauskis, E., & Mouazen, A. M. **2025**. Field-based hyperspectral imaging for detection and spatial mapping of fusarium head blight in wheat. *European Journal of Agronomy*, 164, 127485. Doi: 10.1016/j.eja.2024.127485. **FWCI: 11.75; CP: 2/7; JQ: Q1; C: 9.**
- [4] Gallardo-Romero, D. J.; **Apolo-Apolo, O.E.**; Martínez Guanter, J.; Pérez-Ruiz, M. **2023**. Multilayer Data and Artificial Intelligence for the Delineation of Homogeneous Management Zones in Maize Cultivation. *Remote Sensing*. 15(12), 3131. Doi: 10.3390/rs15123131. **FWCI: 1.47; CP: 2/4; JQ: Q1; C: 2.**
- [5] Casto-Valdecantos, P.; **Apolo-Apolo, O.E.**; Egea, G.; Pérez-Ruiz, M. **2022**. Leaf area index estimations by deep learning models using RGB images and data fusion in maize. *Precision Agriculture*. 23, pp.1949-1966. Doi: 10.1007/s11119-022-09940-0. **FWCI: 5.73; CP: 2/4; JQ: Q1; C: 46.**
- [6] Appeltans, S.; **Apolo-Apolo, O.E.**; Pieters, J.; Pérez-Ruiz, M.; Mouazen, A.M. **2021**. The Automation of Hyperspectral Training Library Construction: A Case Study for Wheat and Potato Crops. 13(23), 4735. *Remote Sensing*. Doi: 10.3390/rs13234735. **FWCI: 0.32; CP: 2/5; JQ: Q1; C: 5.**
- [7] **Apolo-Apolo, O.E.**; Pérez-Ruiz, M.; Martínez-Guanter, J.; Egea, G. **2020**. Design and assessment of new artificial reference surfaces for real time monitoring of crop water stress index in maize.

Agricultural Water Management. 240: 106304. Doi: 10.1016/j.agwat.2020.106304. FWCI: 0.73; CP: (1/4); JQ: Q1; C: 18.

[8] **Apolo-Apolo, O.E.**; Pérez-Ruiz, M.; Martínez-Guanter, J.; Valente, J. **2020**. A Cloud-Based Environment for Generating Yield Estimation Maps From Apple Orchards Using UAV Imagery and a Deep Learning Techniques. *Front. Plant Sci.* 11:1086. Doi: 10.3389/fpls.2020.01086. FWCI: 10.71; CP: 1/4; JQ: Q1; C: 230.

[9] **Apolo-Apolo, O.E.**; Martínez-Guanter, J.; Egea, G.; Pérez-Ruiz, M.; Raja, P. **2020**. Deep learning techniques for estimation of the yield and size of citrus fruits using a UAV. *European Journal of Agronomy*. 115: 26030. Doi: 10.1016/j.eja.2020.126030. FWCI: 10.71; CP: 1/4; JQ: Q1; C: 230.

[10] **Apolo-Apolo, O.E.**; Pérez-Ruiz, M.; Martínez-Guanter, J.; Egea, G. **2020**. A Mixed Data-Based Deep Neural Network to Estimate Leaf Area Index in Wheat Breeding Trials. *Agronomy*. 10(2), 175. Doi: 10.3390/agronomy10020175. FWCI: 2.82; CP: 1/4; JQ: Q1; C: 40.

C.2. Congress

Abbreviations: OP: Oral Presentation; IT: Invited Talk; CP: Author Position / Total Authors.

[1] **Apolo-Apolo, O.E.**; Ben Somers; Stef Lhermitte; Advancing ecosystem modelling: integrating remote sensing and generative AI for predicting climate response. Living Planet Symposium 2025: From Observation to Climate Action and Sustainability for Earth. **2025**. Viena, Austria. OP; CP: 1/3.

[2] **Apolo-Apolo, O.E.**; Transforming the Ecosystem: AI and Hyperspectral Imaging. Reshaping the Frontier of Food Production. 12th Iberian Agroengineering Congress. **2023**. Seville, Spain. IT; CP: 1/1.

[3] Gallardo-Romero, D.J.; **Apolo-Apolo, O.E.**; Pérez-Ruiz, M. Estimating Optimal Harvest Time and Yield in Tomatoes Using Deep Learning Techniques: A Preliminary Study. IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor). **2023**. Pisa, Italy. OP; CP: 2/3.

[4] Gallardo-Romero, D.J.; **Apolo-Apolo, O.E.**; Martínez-Guanter, J.; Pérez-Ruiz, M. Multilayer data and artificial intelligence for the delineation of corn management zones. 14th European Conference on Precision Agriculture. **2023**. Bologna, Italy. OP; CP: 2/4.

[5] **Apolo-Apolo, O.E.**; Almoujahed, M.B.; Whetton L., R.; et al; Mouazen M., A. Modelling and mapping of Fusarium head blight (FHB) in wheat using hyperspectral data under field conditions. 15th IUPAC International Congress of Crop Protection Chemistry. **2023**. New Delhi, India. OP; CP: 1/9.

[6] Rodríguez-Vázquez, J.N.; Egea, G.; **Apolo-Apolo, O.E.**; Castro-Valdecantos, P.; Pérez-Ruiz, M. Determination of wheat rust severity using hyperspectral imagery and 3D plant reconstruction modelling. 13th European Conference on P. Agriculture. **2021**. Budapest, Hungría. OP; CP: 3/5.

[7] **Apolo-Apolo, O. E.**; Martínez Guanter, J.; Andújar Sánchez, D.; Reiser, D.; Pérez-Ruiz, M. Deep-learning-based detection of in-field tomatoes using a terrestrial mobile platform. 13th European Conference on Precision Agriculture. **2021**. Budapest, Hungría. OP. CP: 1/5.

[8] De Groot, J.; **Apolo-Apolo, O. E.**; Valente, J.; Development of a UAV image Dataset for Cauliflowers Ripeness Classification with Deep Learning. 16th European Conference on Computer Vision. 2020. Glasgow, UK. OP. CP: 2/3.

[9] Martínez-Guanter, J.; Egea, G.; Pérez-Ruiz, M.; **Apolo-Apolo, O. E.** Estimation of the leaf area index in maize based on UAV imagery using deep learning techniques. 12th European Conference on Precision Agriculture. **2019**. Bordeaux-Montpellier, France. OP. CP: 4/4.

[10] **Apolo-Apolo, O. E.**, Egea, Gregorio, Pérez Ruiz, Manuel, Martinez Guanter, Jorge. Design of a portable sensor suite for real-time monitoring of crop water stress index in maize breeding plots. 12th European Conference on Precision Agriculture. Bordeaux-Montpellier, France. **2019**. OP. CP:1/4.

C.3. Research projects

Abbreviations: FB: Funded by; IP: Principal Investigator; I: Investigator; FP: Financing period.

[1] **Title:** Deepfaking ecosystem response to climate extremes (DEEPVEEG); **Reference:** STG/22/055; **FB:** Federaal Wetenschapsbeleid; **FP:** October 2022-September 2025; **IP:** Stef Lhermitte; **Budget:** 225.000€; **Call:** AARDOBSERVATIE STEREO IV; **Role:** I participated in this project as an **Investigator** during my postdoctoral work at KU Leuven. I contributed as a researcher by developing cGAN models using satellite data (e.g., Sentinel-2) and weather data (e.g., ERA5) to analyze ecosystem responses to climate extremes. As a result of this work, the oral communication [1] cited in this CV was presented, and three peer-reviewed journal papers are currently in preparation. I also supervised four MSc theses related to this project.

[2] **Title:** Potential of selective harvest based on mycotoxins content assessment in cereal crops (POSHMyCo). **Reference:** Agreement N° 862665 ICT-AGRI-FOOD; **FB:** European Union's Horizon 2020 research and innovation programme. **FP:** January 2021 - March 2024. **IP:** Abdul M. Mouazen; **Budget:** 741.900 €; **Call:** ERA-NET COFUND ICT-AGRI-FOOD 2019. **Role:** I participated in this project as an **Investigator** during my postdoctoral work at Ghent University, where I led the technical components, including fieldwork, AI algorithm development, and data analysis. This work resulted in the oral presentations [3] and the peer-reviewed publications [1] and [3] cited in this CV.

[3] **Title:** Datos Multicapa e Inteligencia Artificial para la Delimitación de Zonas de Manejo Homogéneo en el Cultivo de Maíz (zonIA); **Reference:** PYC20 RE 086 US; **FB:** Junta de Andalucía. **FP:** December 2021-April 2023. **PI:** Manuel Pérez-Ruiz; **Budget:** 79.862,50€; **Call:** Subvenciones de Apoyo a la Investigación (Autonómico); **Role:** I participated in the **Work Team**, contributing to the development of the AI algorithms required to meet the project's objectives. This work led to the publication of a peer-reviewed paper [4] and the oral presentations [3] and [4] cited in this CV.

[4] **Title:** Monitorización en tiempo real del rendimiento y calidad en olivar superintensivo basado en Deep Learning; **Reference:** CEI-15; **FB:** Junta de Andalucía; **FP:** March 2022-September 2021; **PI:** Manuel Pérez-Ruiz; **Budget:** 43.189,30€; **Call:** Subvenciones de Apoyo a la Investigación (Autonómico); **Role:** I participated in the **Work Team** by conducting the experiments and developing a deep learning algorithm for olive detection, which serves as a benchmark for yield monitoring in olive orchards. This work resulted in the publication of a book chapter in *AI Knowledge Transfer from the University to Society*.

[5] **Title:** LiDAR and hyperspectral measurements for the determination of wheat rust severity and its impact on crop yield (ROBIG-US). **Reference:** US-1263678; **FB:** Junta de Andalucía; **FP:** February 2020-January 2022; **PI:** Manuel Pérez-Ruiz & Gregorio Egea; **Budget:** 89.900€; **Call:** Proyectos I+D+i FEDER Andalucía 2014-2020; **Role:** I participated in this project as an **Investigator**, focusing on the experimental component, including data collection and the development of algorithms to assess wheat rust severity. This work resulted in the publication of the peer-reviewed papers [4] and [6], as well as conference oral presentations, specifically [6], cited in this CV.

[6] **Title:** Production estimation and integrated management in fruit trees using drones and artificial vision (DRONFRUIT); **Reference:** GOP2I-SE-16-0048; **FB:** Junta de Andalucía; **FP:** June 2018-June 2020; **PI:** Gregorio Egea Cegarra. **Budget:** 113.668,49 €. **Call:** European Innovation Association (AEI) Task Forces for Agricultural Productivity and Sustainability. **Role:** I participated in this project as an **Investigator**, contributing to the experimental design and the development of AI models for yield prediction. This work resulted in the publication of the peer-reviewed papers [8] and [9], as well as the conference oral presentations [8] and [7] cited in this CV.

[7] **Title:** Advanced Precision Agriculture Service of the Andalusian Agricultural Association. (Smart AG Services); **Reference:** GOP2I-SE-16-0042; **FB:** Junta de Andalucía; **FP:** January 2018-December 2019; **PI:** Manuel Pérez-Ruiz & Gregorio Egea Cegarra; **Budget:** 266.209,66 €; **Call:** Grupo operativo; **Role:** I participated as an **Investigator**, focusing on the installation and validation of an IoT sensor system for monitoring crop water status. Additionally, I contributed to the development of methods to enhance the interpretability of the results.

[8] **Title:** Caracterización de la Respuesta Fenotípica del Trigo y el Maíz al Déficit Hídrico Mediante Tecnologías de Detección Remota y Análisis de Datos (PRESSING); **Reference:** AGL2016-78964-R; **FB:** Ministerio de Economía y Competitividad; **FP:** December 2016-December 2020; **PI:** Manuel Pérez-Ruiz & Gregorio Egea; **Budget:** 108.900€; **Call:** Plan Estatal 2013-2016 Retos-Proyectos I+D+I; **Role:** I participated in this project as an **Investigator** and as part of my PhD research, carrying out field data collection, sensor installation, and experimental design. This work led to the publication of the peer-reviewed papers [5], [7], and [10], and to the conference presentations [8], [9], and [10] cited in this CV.

C.4. Contracts, technological or transfer merits

Abbreviations: FB: Funded by; IP: Principal Investigator; FP: Financing period.

[1] **Title:** In Company Tragsatec: Geoprocesamiento de procesos en Python; **FB:** Campus Cámara de Sevilla SAU; **FP:** November 2025; **PI:** Orly Enrique Apolo-Apolo; **Budget:** 1625€.

[2] **Title:** Expert course in artificial intelligence and robotics for the agriculture sector; **FB:** Colombian School of Engineering Julio Garavito; **FP:** June 2023 **PI:** Orly Enrique Apolo-Apolo; **Budget:** 1000€.

[3] **Title:** Proyecto de capacitación de Ingenieros Técnicos Agrícolas. Herramientas cartográficas con QGIS, python para Sistemas de Información Geográfica y DataScience. **FB:** Campus Cámara de Sevilla SAU; **FP:** April 2021 **PI:** Orly Enrique Apolo-Apolo; **Budget:** 2700€.